Overview
The ability to design and implement an information system solution to effectively manage vast amounts of information is a valuable skill that leads to the success of many business entities today. The use of technology to develop these information systems plays a crucial role in a business' ability to compete in today's business environment. This event provides recognition for FBLA members who demonstrate an understanding of and ability to apply these skills.

This is a team event composed of two or three members. This event consists of two parts: an objective test and a performance. The objective test is taken collaboratively by the team members, and the top ten (10) teams scoring the highest on the objective test will advance to the final round and participate in the performance component.

Competencies
The topics listed below are prioritized, listing the most important content areas of the tests.

Objective Test Competencies
- Systems analysis and design—systems development life cycle
- Database management and modeling concepts
- Object-oriented analysis and design
- User interfaces
- System controls
- Defining system and business requirements

Performance Competencies
- Human resource management
- Financial management
- Marketing management
- Information systems management

Procedures/Tips
- Review the event guidelines in the *CMH* since guidelines may change.
- Review the Competitive Events Tips in the front of the book.

Management Information Systems Sample Questions

1. An information system
   a. is always Internet based.
   b. is an organized combination of people, hardware, software, communications, networks, and database resources.
   c. is a system that deletes and corrupts crucial information in an organization.
   d. is always computer based.

2. Users of a management information system include
   a. all of the answers are correct
   b. entry-level employees.
   c. managers.
   d. persons and organizations in a business's environment.

3. Which of the following steps is not part of the software development life cycle?
   a. implementation
   b. engineering
   c. design
   d. analysis

4. What act provides policy to ensure public access to federal government information?
   a. Privacy Act
   b. Federal Records Act
   c. Paperwork Reduction Act
   d. Freedom of Information Act

5. Which of the following are ways we manage information in support of information resources?
   a. security and privacy
   b. storage and privacy
   c. storage and disposition
   d. security and disposition

6. An integrated collection of stored data that is centrally managed and controlled is also known as a
   a. table.
   b. spreadsheet.
   c. query.
   d. database.

7. During what phase of the software development life cycle is a system finalized and installed?
   a. support
   b. development
   c. implementation
   d. planning

8. The phase of the software development life cycle that includes defining system requirements, prioritizing requirements, and generating alternatives is the
   a. analysis phase.
   b. planning phase.
   c. implementation phase.
   d. support phase.

9. CASE tools help an analyst create important system models by checking for completeness and compatibility with other systems. The acronym CASE stands for
   a. computer analysis structuring entity.
   b. computer-automated structured engineering.
   c. compatibility analysis system entity.
   d. computer-aided system engineering.

10. A graphical model of data needed by a system, including things about which information is stored and the relationships among them, produced in structured analysis and information engineering is called a
    a. structured design model.
    b. data flow diagram.
    c. entity relationship diagram.
    d. class diagram.

11. The Unified Process development methodology is a tool developed by
    a. Rationale.
    b. Oracle.
    c. Microsoft.
    d. PeopleSoft.

12. A popular project management software application is
    a. Corel Analysis and Design.
    b. Oracle Management Design.
    c. Microsoft Design.
    d. Microsoft Project.

13. Viewing project tasks and activities over a calendar utilizes a
    a. PERT/CPM chart.
    b. Dataflow diagram.
    c. Constructive Cost Model.
    d. Gantt chart.

14. Iteration means
    a. activities are done once and not repeated.
    b. activities are repeated and refined each time they are repeated.
    c. activities are repeated in exactly the same steps each time.
    d. activities are done one and only revisited during system modifications.

15. Interviewing users to gather system and business requirements is part of what phase of the software development life cycle?
    a. analysis
    b. implementation
    c. testing and support
    d. design

16. The acronym CRUD stands for
    a. class, report, user, data.
    b. chart, rationalize, understand, design.
    c. cost, return, utilization, depreciation.
    d. create, read, update, delete.

17. Which of the following is not an example of a data flow diagram?
    a. context diagram
    b. process flow diagram
    c. DFD fragment diagram
    d. event-partitioned system model

18. A repository for definitions of data flows, data sorts, and data elements is called a
    a. data dictionary.
    b. data summary.
    c. CASE tool.
    d. decision tree.

19. Which of the following is **not** a rule that should be followed when designing user interfaces?
    a. use of shortcuts
    b. creative and original design
    c. simple error handling
    d. consistent design

20. When designing user interfaces, the process for documenting dialog designs by showing a sequence of sketches of the display screens is called
    a. sequencing.
    b. storyboarding.
    c. prototyping.
    d. diagramming.

21. The data entry control that allows a user to select one option from a group is called
    a. text box.
    b. none of the answers are correct.
    c. radio button.
    d. checkbox.

22. Which of the following diagrams is not a major component of the object-oriented approach to system development?
    a. state chart diagram
    b. system sequence diagram
    c. context diagram
    d. use case diagram

23. Benefit of the object-oriented approach to systems development include
    a. intuitive design.
    b. all of the answers are correct.
    c. use of objects instead of complex procedures.
    d. reuse of objects.

24. What is the term that describes a consistent relational database state in which every foreign key value also exists as a primary key value?
    a. referential integrity
    b. normalization
    c. schema
    d. functional dependency

25. Viewing project tasks and activities over a series of slides utilizes a
    a. PERT/CPM chart.
    b. Dataflow diagram.
    c. Constructive Cost Model.
    d. Gantt chart.

26. A _____ is responsible for ensuring that system's networking requirements is compatible with a business's existing network.
    a. management analyst
    b. network analyst
    c. systems analyst
    d. business analyst

27. The process of normalization helps monitor's a database's quality by reducing data
    a. redundancy.
    b. delays.
    c. logs.
    d. efficiency.

28. The _____ approach may be used when the project development schedule is shortened and a well-trained and experienced staff is involved.
    a. slow-and-methodical
    b. rapid application development
    c. seminar
    d. conference

29. Which of the following is **not** required of storyboarding?
    a. pictures
    b. sequences
    c. subject/topic
    d. organization

30. A _____ needs to understand how what business functions an organization performs.
    a. management analyst
    b. network analyst
    c. systems analyst
    d. business analyst

Performance Indicators

- System appropriate for size of business
- System solution is feasible and realistic given specified time frame
- Technology is currently available
- Future needs are considered
- Information security issues are addressed
- Meets the needs of the company
- Demonstration of ability to effectively answer questions
- Described the situation
- Issued a solution or recommendation
- Used correct terminology
- Effectiveness of the strategy presented
- Voice quality and diction
- Professional and poised
- Self-confidence and assertiveness

Background Information

The Vinings Community Library currently maintains a manual system for the following functions:

- Cataloguing
- Checking books out and back in
- Book inventory
- Book requests and waiting lists
- Keeping track of late fees
- Customer information

The library recently received a large grant from a local foundation to update its manual system. The library would like the system designed, developed, and implemented within an 8-month period. In order to develop a Request for Proposal (RFP), the library has requested that your team develop a basic system and process functionality design. Here is the data and process for each function of the library:

Library information:
The library currently has an inventory of 25,000 books. It is has 2,000 registered customers, with approximately 800 books being checked out at any given time.

Customer information:
Customers currently fill out a card to provide name, phone number, address, age, and parent's information (if applicable). If the customer incurs late fees, these are manually noted on the customer card. Once a customer fills out the customer information card, they receive a library card with a unique number. The customer information card is then stored in customer card file box.

Cataloguing process:
For each book that a library receives, a card is typed up that is stored in the card catalog system. It contains information about the book such as title, author, classification, dewey decimal number, number of copies, brief description, restrictions. This card is then stored in the card catalog file.

Book check-out process:
Each book has a check-out card in the back flap. The check-out card lists the book title, author, copy number (in case there are multiple copies of the same book), and a reference number. There is also a table where the library staff records the customer's name, check-out date and due date. When a book is checked out, the customer's information is recorded on the card and a bookmark is given to the customer with the book with the due date. The check-out card is then kept in the check-out card file box in alphabetical order by title. If a customer cannot find a book on the library shelves and inquires with the staff if a book is available, the staff must go through the check-out card file box to see if a book is checked out and when it is due to be turned in.

Overdue books notification process:
Every other week, the staff goes through the check-out cards to determine any books overdue. If a book is overdue, the staff will pull the card. The staff then has to look up the customer information from the customer card file to contact the customer.

Book check-in process:
When a book is checked in, the library staff has to get the check-out card from the card file and crosses out the customer's information on the check-out card. If the card is full, then a new card must be created. The check-out card is then returned to the book and the book is placed on the reshelving cart. If a book is overdue when it is checked in, the library staff has to manually calculate the late fees.

Reservation process:
Occasionally, customers and other libraries will want to reserve a book that is currently checked out. When someone makes a reservation, the library staff records that customer's information on a sticky note and attaches that sticky note to the check-out card for that book. When the book is checked in, the staff sees the sticky note and sets that book aside on the reservations cart if it is reserved by another customer. The books on the reservations cart should be kept in alphabetical order by title. If another library requests the book, the check-out card is then updated with the borrowing library's information and the book is sent to the requesting library.

Current issues:

- As a result of the manual system, it often takes much effort to determine which books are checked out and which are in the current inventory. It is a labor-intensive process that relies on the accuracy of the employees to update the records. If they fail to update the records properly, then the information the library has is out of date.
- The library staff often has to check multiple card file boxes to get all of the information.
- Sometimes the sticky notes (for reservations) fall off the check-out cards and the reservation is then lost.
- The current process relies on the accuracy of the library staff in following the process. There are multiple places for a breakdown of the process and confusion if the library staff forgets a certain step.

Case Study
The library staff would like a system developed that meets the following goals:

- Puts cataloguing system online so that customers can look up information about books at the library as well as determine if the book is checked out and if so, when it is expected to be returned.
- Calculates late fees automatically at check in
- Limits updating access about books to the library staff only (customers cannot update information in the system)
- Improves process of identifying books that are overdue and customer information for overdue books
- Improves process for library staff to determine which books are checked out
- Allows customers to fill out customer information, make reservations, and even request books for the library to purchase online
- Streamlines the book check-out and check-in process so that minimal data has to be entered by library staff
- Maintains a history of the books that a customer checks out and well as a history of which customers checked out a certain book
- System should be designed so that simple maintenance issues can be handled by the library staff

Other Useful Information
In addition to designing a system/process that meets the above goals, you should also consider the following issues:

- What information security issues need to be addressed?
- What other processes or enhancements would be available by your proposed system?
- What system maintenance will be required by library staff or outside contractors?
- What enhanced reporting features will be available to improve current operations for library managers?
- How will system access be controlled?
- How will the system be tested and implemented and will this result in any time that the library is closed to convert to the new system.
- How will the library staff be trained on the new system?
- What equipment will be required to implement the new system?

Introduction to Business Communication Answer Key

1) A	11) A	21) B
2) A	12) D	22) B
3) B	13) C	23) A
4) A	14) D	24) B
5) A	15) A	25) A
6) A	16) B	26) B
7) B	17) B	27) A
8) A	18) B	28) C
9) B	19) B	29) B
10) B	20) A	30) B

Introduction to Parliamentary Procedure Answer Key

1) C	11) B	21) A
2) B	12) C	22) B
3) A	13) C	23) B
4) B	14) C	24) B
5) A	15) B	25) B
6) C	16) C	26) A
7) A	17) B	27) A
8) B	18) A	28) B
9) C	19) B	29) A
10) C	20) A	30) A

Introduction to Technology Concepts Answer Key

1) B	11) A	21) D
2) D	12) A	22) A
3) B	13) A	23) B
4) A	14) B	24) A
5) C	15) C	25) B
6) C	16) D	26) A
7) A	17) B	27) B
8) A	18) D	28) C
9) D	19) A	29) C
10) D	20) A	30) B

Management Decision Making Answer Key

1) B	11) C	21) C
2) D	12) D	22) D
3) A	13) A	23) A
4) A	14) C	24) A
5) C	15) C	25) A
6) D	16) A	26) B
7) B	17) A	27) C
8) A	18) D	28) B
9) D	19) A	29) D
10) A	20) A	30) A

Management Information Systems Answer Key

1) B	11) A	21) C
2) A	12) D	22) C
3) B	13) D	23) B
4) D	14) B	24) A
5) A	15) A	25) B
6) D	16) D	26) B
7) C	17) B	27) A
8) A	18) A	28) B
9) D	19) B	29) A
10) C	20) B	30) B